

The Closed Form of the Period

the beta integral, the phase lattice, and equivariance

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The «Dynamic Principle» Program — the hodge-laboratory

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1 Setting

The periods of the Fermat curve — integrals of holomorphic forms over cycles — admit a closed form: the transcendental part (a beta integral) and the algebraic part (a root of unity) separate explicitly. This separation is the technical foundation of the whole Γ -normalization.

Lemma 1 (basis of holomorphic forms). *The forms $\eta_{i,j} = \frac{x^i y^j dx}{y^{N-1}}$, $i, j \geq 0, i+j \leq N-3$, are holomorphic on C_N ; their number is g , and they form a basis of $H^0(C_N, \Omega^1)$. Each $\eta_{i,j}$ is an eigenform of the $\mu_N \times \mu_N$ -action with character $(a, b) = (i+1, j+1)$.*

Lemma 2 (closed form of the period). *In the coordinate $t = x^N$, $\eta_{i,j} = \frac{1}{N} t^{\frac{a}{N}-1} (1-t)^{\frac{b}{N}-1} dt$. For a lift γ with winding data $(r, s) \in (\mathbb{Z}/N)^2$ the period is*

$$P_{a,b}(r, s) = \frac{1}{N} \zeta_N^{ra+sb} \Omega_{a,b}, \quad \Omega_{a,b} = \frac{\Gamma(\frac{a}{N})\Gamma(\frac{b}{N})}{\Gamma(\frac{a+b}{N})} > 0,$$

and the image of the period functional lies in $\frac{\Omega_{a,b}}{N} \mathbb{Z}[\zeta_N]$.

Proof. The substitution $t = x^N$ gives $dt = Nx^{N-1}dx$ and $x^i dx = \frac{1}{N} t^{\frac{i+1}{N}-1} dt$; and $y^j/y^{N-1} = (1-t)^{\frac{j+1}{N}-1}$ since $y^N = 1-t$. Each loop around $t = 0$ multiplies the branch $t^{1/N}$ by ζ_N — the integral by ζ_N^a ; a loop around $t = 1$ gives ζ_N^b . A path with data (r, s) carries ζ_N^{ra+sb} on the reference integral $\frac{1}{N} B(a/N, b/N)$. Holomorphy: at $(\zeta, 0)$, $dx \sim -\zeta y^{N-1} dy$; at infinity in $u = X/Y, v = Z/Y$ the form is $u^i(v du - u dv)v^{N-3-i-j}$, regular at $i+j \leq N-3$. Independence: forms of distinct characters are independent (projection operators). Lattice: every cycle is an integer combination of closed chains; the edge integral is a difference of two closed-form values. $\square$

Corollary 3 (equivariance). *Translating a cycle by (ζ^u, ζ^v) multiplies the period of the eigenform (a, b) by ζ_N^{ua+vb} .*

Protocol data: numbers

$N = 15$, $(a, b) = (1, 1)$: $\Omega_{1,1} = 246.897\dots$; $N = 15$, $(7, 8)$: $\Omega_{7,8} = \pi/\sin 84^\circ = 3.16298\dots$ — a pure fractional share of π . Certificate B (V2): maximal relative deviation from independent tanh-sinh over 69 tests — $3.9 \cdot 10^{-36}$; phases (V3) — deviation 0.0.

Summary of results: summary

The closed form is proved by substitution and branch analysis; equivariance is a direct corollary. The phase is fixed explicitly — no “global phase” is sought or postulated.

Verification: reproduction

`python3 laboratory.py` → “Stand $N=15/30 \rightarrow V2/V3/V4$ ”; the experiment designer accepts arbitrary N , (a, b) , (r, s) and prints the closed form, the independent integral, and the verdict.

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